The Federal University of Technology, Akure
School of Engineering and Engineering Technology
Department of Information and Communication Technology
First Semester Examinations 2023/2024 Session
ICT 509, Data Structures and Algorithms (DSA); Date: 27th of May, 2024: 3 HOURS.

General Instruction: Answer ALL QUESTIONS in Section A and ONE (1) QUESTION EACH from Section B, C and D.

SECTION A

1. The efficiency of DSA is always measured in terms of TWO parameters……………………………

2. List TWO examples of DS which we studied in this course this semester……………………………

3. List TWO methods that can be used to determine the time complexity of any DSA……………

4. Which of the methods in (3) is the most used today……………………………………………

5. What is the major parameter used in the application of the method chosen in (4) ……………

6. What is the upper bound time complexity of a nested loop……………………………………

7. What type of DS is a graph…………………………………………………………………………

8. A graph without loops and multiple edges between two nodes is known as……………………

9. A directed graph has two types of degrees, list them……………………………………………

10. The condition used to stop a recursive algorithm is known as…………………………………

11. To divide a problem into smaller sub-problems in a recursive algorithm, you need a what……

12. For a complete undirected graph, state the formular to calculate its total spanning tree………

13. For a complete graph, state the formular to determine the maximum number of edges that can be removed to construct a spanning tree from it……………………………………………………

14. Breadth-First-Search (BFS) algorithm uses which data structure for its graph traversal………

15. On what principle does this data structure provided in (14) operates…………………………

16. Depth-First-Search (DFS) algorithm uses which data structure for its graph traversal…………

17. On what principle does this data structure provided in (16) operates…………………………

18. BFS and DFS have the same time complexity, state it……………………………………………

For questions 19 to 21, provide either BFS or DFS as an appropriate answer.

19. Facebook and LinkedIn friend suggestion are based on ……………………………………traversal

20. Solve maze or sudoku having only one solution…………………………………………………

21. Detect a cycle in a graph……………………………………………………………………………

Consider the C function given below, provide either TRUE or FALSE as an appropriate answer for questions 22 to 25.

22. The function returns 0 for all values of j……………………………………………………………

23. The function prints the string ***'something'*** for all values of j…………………………………

24. The function returns 0 when j = 50…………………………………………………………………

25. The function will run into an infinite loop when j = 50

```
int f(int j)
{
   static int i = 50;
   int k;
   if (i = = j)
   {
     print f ('something');
     k = f(i)
     return 0;
   }
   else return 0;
}
```

SECTION B

1. (a). i). What is ***Data Structure*** (DS), and how does your understanding of ***Data Structure and Algorithm*** (DSA) contributes to making you become a better programmer. **(4 marks)**

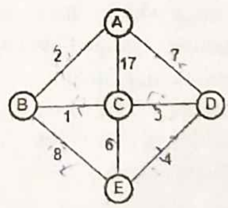
   ii). State Five (5) differences between ***Data*** and ***Information***. **($2\frac{1}{2}$ marks)**

   (b). i). Briefly explain what the ***Big O notation*** means in the context of analyzing Time Complexity in DSA. **(2 marks)**

   ii). Explain briefly the difference between the big O notations $O(1)$, $O(n)$, and $O(n^2)$. Provide simple code snippets to support your explanations. **(6 marks)**

   (c). i). What do you understand by ***Recursion*** and how does it differ from ***Iteration***? Use a simple code snippet to illustrate your discussion. **(5 marks)**

   ii). Consider the graph given below and construct its ***adjacency matrix***. Apply ***Kruskal's algorithm*** to find its ***Minimum Spanning Tree (MST)*** with its cost. **($5\frac{1}{2}$ marks)**



2. (a). i). Provide a precise definition for the term ***Algorithm***. **(3 marks)**

   ii). State the ***five (5)*** basic structures of an Algorithm. **($2\frac{1}{2}$ marks)**

   (b). i). Arrange the following functions in decreasing order of time complexity in DSA: 2^n, n^2, 2^{2n}, 2^{logn}, n, $n!$, $nlogn$, $log(n!)$, 1, $loglogn$, $(logn)^{1/2}$. **($5\frac{1}{2}$ marks)**

   ii). How does the Big O notation handle constant factors and lower-order terms in the analysis of algorithmic complexity? **(2 marks)**

   (c). i). Using a recursive algorithm, write a simple program that prints linearly from 1 to N =

4, and analyze and visualize the output of your program using a ***Recursive Tree and Stack***. **(8 marks)**

ii). What do you understand by a ***Complete graph*** and a ***Strongly connected graph***? **(4 marks)**

SECTION C

3. (a). i). List THREE ways in which an ***Algorithm*** differs from a ***Program*** in DSA. **(3 marks)**

ii). List the steps involved in the development of an Algorithm. **(4 marks)**

(b). i). Describe briefly the significance of ***Asymptotic Analysis*** in evaluating the efficiency and scalability of algorithms. **(3 marks)**

ii). Write two programs that linearly print out the sum of the first N natural numbers. The first program should have ***O(n)*** while the second should have ***O(1)*** time complexity respectively. **(5 marks)**

(c). i). Discuss the concept of ***"Stack overflow"*** in the context of recursive algorithms. Use a simple code snippet to illustrate your discussion. **(4 marks)**

ii). What do you understand by the ***"minimum spanning tree"*** (MST) of a graph? State the relationship between the ***Vertex (V)*** and ***Edge (E)*** of a graph and its MST. **(6 marks)**

4. (a). i). Briefly describe the concept of ***Abstract Data Type*** (ADT) in the context of DSA. **(3 marks)**

ii). What are the advantages of ADT in DSA? **(3 marks)**

(b). With the aid of simple code snippets, what is the time complexity of the following block of a program?

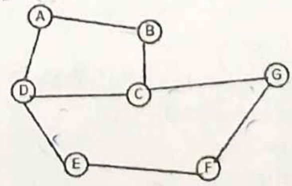
i). Loops. **(3 marks)**

ii). Nested loops **(3 marks)**

iii). If-then-else statements. **(3 marks)**

(c). i). Provide the algorithm for the ***Breadth-First-Search*** (BFS) graph traversal method. **(4 marks)**

ii). Consider the ***undirected graph*** as shown below, and determine **ONE** order of its traversal using the BFS algorithm. **(6 marks)**



SECTION D

5. (a). i). Using a Tree Diagram, state the classifications of DSA. **(3 marks)**

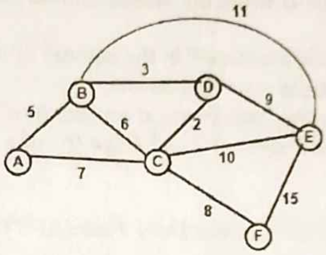
ii). Briefly discuss THREE advantages does DSA provide in the writing of Algorithms? **(3 marks)**

(b). Analyze how the simple code below has a time complexity of $O(\log_2 N)$ **(8 marks)**

```
for (i = 1;  i <= N;)
{
    i = i * 2;
}
```

(c). i). State the step-by-step ***Prim's algorithm*** in finding the ***minimum spanning tree*** of an ***undirected graph.*** **(3 marks)**

ii). Use Prim's algorithm to determine the minimum spanning tree of the below graph. **(8 marks)**



6. (a). State what the following means in the organization of DSA

i). *Entity* and *Attributes.* **(2 marks)**

ii). *Entity set.* **(2 marks)**

iii). *Fields*, *Records,* and *Files*. **(3 marks)**

(b). i). State the formal definition of the *Big O Notation*. **(2 marks)**

ii). Using the definition stated in *6.(b).(i),* what is the time complexity of $f(n) = 5n^2 + 4$ and $f(n) = 4n + 3$. **(4 marks)**

(c). Consider the following information about the directed graph as defined below for a traffic flow in Akure town:

G(V, E);

V(G) = {1, 2, 3, 4, 5, 6};

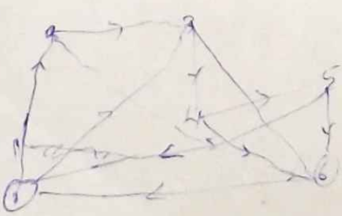
E(G) = {(1, 2), (2, 3), (3, 4), (4, 5), (5,1), (5, 6), (2, 6), (1, 6), (3, 6), (4, 6), (2, 3)}

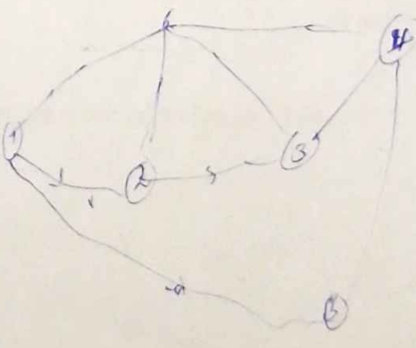
Based on this information, answer the following:

i). Construct its *graph* structure. **(6 marks)**

ii). Compute its *adjacency matrix*. **(3 marks)**

iii). Compute the *In-degree* and *Out-of-degree* of each vertex. **(3 marks)**





THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE

SCHOOL OF ELECTRICAL SYSTEMS ENGINEERING

DEPARTMENT OF INFORMATION AND COMMUNICATION ENGINEERING

B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION

COURSE CODE: ICT 509 **COURSE TITLE:** Data Structure and Algorithm

UNITS: 2 **TIME ALLOWED:** 3 Hours

Instruction: Answer ALL QUESTIONS in Section A and ONE QUESTION EACH from Section B, C and D.

SECTION A

1. Breadth-First Search (BFS) and Depth-First Search (DFS) are primarily used for __________ (directed/undirected/both) graphs and can work on __________ (weighted/unweighted/both) graphs.
2. Dijkstra's Algorithm is used to find the shortest path in __________ (directed/undirected/both) graphs but does not work if the graph contains __________ (negative/positive) weight edges.
3. Bellman-Ford Algorithm can handle graphs with __________ (negative/positive/both) weight edges and works on __________ (directed/undirected/both) graphs.
4. Kruskal's Algorithm is used to find the Minimum Spanning Tree (MST) and is only applicable to __________ (directed/undirected/both) graphs.
5. Prim's Algorithm, like Kruskal's, is used to compute an MST but only works for __________ (directed/undirected/both) graphs and must have __________ (weighted/unweighted/both) edges.
6. Which algorithm(s) can be used for shortest path weighted graph problems? __________ (BFS/DFS/Dijkstra/Bellman-Ford/Kruskal/Prim)
7. Which algorithm(s) can be used to detect negative weight cycles? __________ (BFS/DFS/Dijkstra/Bellman-Ford/Kruskal/Prim)
8. Which algorithm(s) build a spanning tree rather than finding a shortest path? __________ (BFS/DFS/Dijkstra/Bellman-Ford/Kruskal/Prim)
9. What is the time complexity of accessing an element in an array using its index? (A) O(n), (B) O(logn) (C) O(1), (D) O(nlogn).
10. What is the time complexity of inserting an element at the beginning of an array? (A) O(1), (B) O(logn) (C) O(n), (D) O(nlogn)
11. What is the best case time complexity for searching an element in an unsorted array? (A) O(1), (B) O(log n), (C) O(n), (D) O(nlogn)
12. If an array has n elements, what is the time complexity of reversing it in place? (A) O(n), (B) O(nlogn), (C) O(1), (D) $O(n^2)$
13. Which of the following best describes Big-O notation? (A) It gives the best-case complexity of an algorithm. (B) It provides the exact runtime of an algorithm. (C) It describes the upper bound of an algorithm's growth rate. (D) It only applies to recursive algorithms.
14. Which notation represents the best-case complexity of an algorithm? (A) O(n) (B) Θ(n) (C) Ω(n) (D) O(1)
15. If an algorithm has a time complexity of $O(n^2)$, what happens when the input size doubles? (A) Execution time doubles (B) Execution time increases fourfold (C) Execution time remains the same (D) Execution time halves.
16. Which of the following complexities grows the slowest? (A) O(n!) (B) $O(n^2)$ (C) O(nlogn) (D) O(logn)
17. If an algorithm has two independent loops, each running O(n), what is the total time complexity? (A) O(n!) (B) $O(n^2)$ (C) O(2n) (D) O(logn)
18. If a complete graph has **V** vertices and **E** edges, the number of spanning trees that can be formed is given by __________.

19. A spanning tree does not contain any ________, meaning there is only one unique path between any two vertices.
20. A graph with **n** vertices will have exactly ________ edges in its spanning tree.
21. Asymptotic analysis is only useful when dealing with small input sizes. **(True/False)**
22. An algorithm with $O(2^n)$ time complexity is more efficient than $O(n^2)$ for large inputs **(True/False)**
23. Deleting an element from an array requires shifting all subsequent elements **(True/False)**
24. Searching for an element in an unsorted array always takes $O(\log N)$ time **(True/False)**
25. You can insert elements at any position in an array in $O(1)$ time **(True/False)**

SECTION B

1. (a). (i). Define a data structure and explain its importance in computer science. *(2 marks)*
 (ii). Using a tree diagram, list out the classification of Data structure. *(2 marks)*
 (iii). What are the primary differences between linear and non-linear data structures. *(3 marks)*
 (b). (i). What does **Big O** mean and why is it very important. *(3 marks)*
 (ii). Why is **O(n!)** considered impractical for large inputs? *(2 marks)*
 (iii). Calculate the **f(n)** and **g(n)** for this for loop expression: $for(i = 3; i <= n; i = i^2)$. *(3 marks)*
 (c). Consider the following information about the directed and weighted graph as defined below for a traffic flow in Akure town:
 G(V, E); **V(G)** = {A, B, C, D, E, F}; **E(G)** = {(A, B, 5), (B, C, 7), (C, D, 10), (D, E, 2), (E, A,1), (E, F, 7), (B, F, 4), (A, F, 5), (C, F, 6), (D, F, 10), (B, C, 4)} Based on this information, answer the following:
 (i). Construct its graph structure and compute its adjacency matrix. *(4 marks)*
 Pending (ii). Assuming the graph structure in **(1.c.i)** is undirected but weighted, re-construct the new graph structure and determine its minimum cost using Prim's Algorithm, show all workings. *(6 marks)*

2. (a). (i). What is an Abstract Data Type (ADT) and how it differs from a data structure. *(2 marks)*
 (ii). List four fundamental operations that can be performed on data structures. *(2 marks)*
 (iii). What is an algorithm? Describe the key characteristics that a good algorithm should have *(3 marks)*
 (b). (i). Briefly state the three types of asymptotic notations and what they are used for? *(3 marks)*
 (ii). Explain why **O(1)** is considered the most efficient time complexity. *(1 mark)*
 (iii). Derive the **f(n)** and **g(n)** for this for loop expression: $for(i = n; i >= 5; i = \sqrt{i})$. *(3 marks)*
 (c). Consider the graph structure of Figure 1 and answer the following questions using the Bellman-Ford Algorithm.
 (i). Show if there's a negative cycle in the graph. *(2 marks)*
 (ii). Determine the minimum distances between the starting point **S** to all other vertexes. *(7 marks)*
 (iii). Identify the shortest path route from **S** to **D** and **S** to **B** based on your calculations. *(2 marks)*

SECTION C

3. (a). (i). Define an **array** as a data structure. Explain the key characteristics that differentiate it from other data structures. *(3 marks)*
 (ii). Explain the concept of **indexing** in arrays and how it affects the time complexity of searching for an element. *(3 marks)*
 (iii). Given the array: A = [12, 5, 8, 20, 7, 15], write the array index positions for each element and determine the value stored at index 3. *(2 marks)*
 (b). (i). Why is asymptotic analysis more useful than measuring actual runtime in seconds? *(2 marks)*
 (ii). What is space complexity, and how does it differ from time complexity. *(2 marks)*
 (iii). Derive the **f(n)** and **g(n)** for this for loop expression: $for(i = n^2; i >= 1; i = i/2)$. *(4 marks)*
 (c). A logistics company is planning the most efficient delivery route between warehouses in a city. The city has six major warehouses labeled **W, X, Y, Z, D,** and **Q.** The company needs to find the shortest path from Warehouse **W** to all other warehouses. The road network between

V[G] = {W, X, Y, Z, D, Q}
E[G] = {(W,X,3), (W,Y,5), (X,Z,6), (Y,Z,6), (Y,D,4) (Z,D,1) (Z,Q,8), (D,Q,3), (X,D,7) (X,Y,2)}

warehouses is represented as a **directed and weighted graph**, where the edge weights indicate the transportation cost in units. The network is defined as follows:
G(V, E); **V(G)** = {W, X, Y, Z, D, Q}; **E(G)** = {(W, X, 3), (W, Y, 5), (X, Z, 6), (Y, Z, 6), (Y, D, 4), (Z, D, 1), (Z, Q, 8), (D, Q, 3), (X, D, 7), (X, Y, 2)}

(i). Draw the **directed weighted graph** based on the given vertex and edge set. Ensure that all edge directions and weights are correctly represented. *(2 marks)*

(ii). Apply **Dijkstra's Algorithm** to determine the shortest distances from Warehouse **W** to all other warehouses. *(4 marks)*

(iii). Identify the shortest path route from W to Q and W to D based on your calculations. *(3 marks)*

4. (a). A university wants to store the scores of 50 students in a programming exam.

i). Suggest an appropriate data structure to store the scores. *(1 mark)*

ii). Write a C-style or Python-style declaration for storing the scores. *(2 marks)*

iii). Write a code that inserts a new score at a specific position efficiently? *(3 marks)*

(b). (i). How does time complexity affect algorithm choice in real-world applications. *(2 marks)*

(ii). Rank the following time complexities from fastest to slowest growth rate: $O(n^2)$, $O(1)$, $O(n \log n)$, $O(n!)$, $O(\log n)$, $O(n)$. $O(1), O(\log n), O(n), O(n\log n), O(n^2), O(n!)$ *(3 marks)*

(iii). Derive the **f(n)** and **g(n)** for this for loop expression: $for(i = 5^n; i >= 5; i = \sqrt[5]{i})$. *(4 marks)*

(c). Consider the graph in **Figure 2** and answer the following questions

(i). Perform a **Depth First Search (DFS) traversal** starting from vertex **A**. Clearly show the step-by-step process of the traversal, including the order in which nodes are visited and the data structure operations involved. Represent the final DFS traversal sequence clearly. *(7 marks)*

(ii). Which type of data structure is used to store and manage the traversal process in the DFS? Explain why this data structure is suitable for DFS. *(3 marks)*

SECTION D

5. (a). (i). Given an undirected graph with 5 vertices and the following edges: (A, B), (A, C), (B, D), (C, E), and (D, E), construct the graph and its corresponding adjacency matrix. *(3 marks)*

(ii). State 2 merits and demerits of using an **adjacency matrix** for graph representation. *(2 marks)*

(iii). What is the difference between a **Sparse graph** and **Simple graph.** *(2 marks)*

(b). (i). Derive the **f(n)** and **g(n)** for this nested for loop expression:

$$for(i = 1; i <= n; i = i^3)$$
$$for(j = n; j >= 1; j = j/2)$$
$$for(k = 1; k <= n; k = k + 4)$$

(7 marks)

(ii). Find the upper bound for this expression: $f(n) = 2n^3 - 2n^2$ *(4 marks)*

(c). A university is planning to connect its main facilities with pedestrian walkways. The facilities include: **A** (Administration Building), **B** (Library), **C** (Engineering Faculty), **D** (Science Faculty), **E** (Hostel Area), **F** (Sports Complex). The Network of constructing the walkways between these facilities is described as follows: **G(V, E)**; **V(G)** = {A, B, C, D, E, F} **E(G)** = {(A, B, 10), (A, C, 25), (A, D, 40), (B, C, 20), (B, E, 35), (C, D, 15), (C, E, 30), (D, E, 10), (D, F, 25), (E, F, 20)}

Using the given information:

(i). Model the problem as an **undirected weighted graph**, where the facilities are nodes and the walkways are edges with given weights. *(2 marks)*

(ii). Determine the minimum cost to connect all the facilities using **Kruskal's Algorithm**. *(5 marks)*

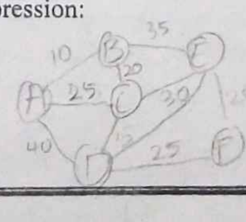
6. (a). (i). Define an Incidence matrix as used on graph data Structure *(2 marks)*

(ii). Given a directed graph with 4 vertices (A, B, C, D) and edges: (A → B), (A → C), (B → D), (C → D), construct the corresponding graph and its incidence matrix. *(3 marks)*

(iii). What is the difference between a **Dense graph** and a **Complete graph.** *(2 marks)*

(b). (i). Derive the **f(n)** and **g(n)** for this nested for loop expression:

$$for(i = n/2; i <= n; i = i++)$$
$$for(j = 1; j + n/2 <= n; j = j++)$$



$for(k = 1; k <= n; k = k * 2)$ *(7 marks)*

(ii). Find the upper bound for this expression: $f(n) = n^4 - 100n^2 + 35$. *(4 marks)*

(c). A university is designing an emergency response system for its campus to ensure quick evacuation and efficient movement of emergency personnel during a crisis. The university is represented as an unweighted graph, where nodes represent key locations on campus and edges represent pathways connecting these locations (all paths are bidirectional and unweighted). The following connectivity data represents the university's movement network:

V(G) = {A, B, C, D, E, F, G, H, I, J, K, L} E(G) = {(A, B), (A, C), (B, D), (B, E), (C, F), (D, G), (E, H), (F, I), (G, J), (H, K), (I, L), (J, K), (K, L), (E, F), (G, H), (D, F), (C, G)} biger

(i). Using the provided data, construct an undirected/unweighted graph representing the university's emergency response network. Clearly label the nodes and edges to reflect the connections between key locations. *(2 marks)*

(ii). Perform a **Breadth First Search (BFS)** traversal starting from the library (node **C**) and construct a tree diagram identifying at least 4 paths of evacuation route to the nearest emergency exit (node **K**). Show all traversal steps, including queue operations. *(5 marks)*

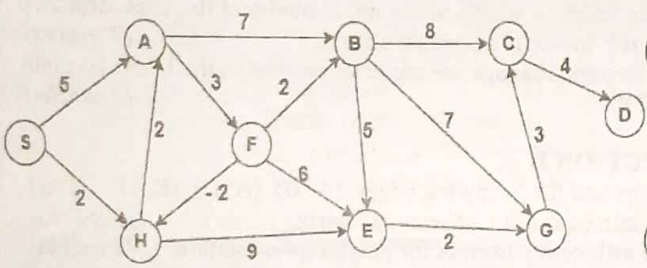


Figure 1

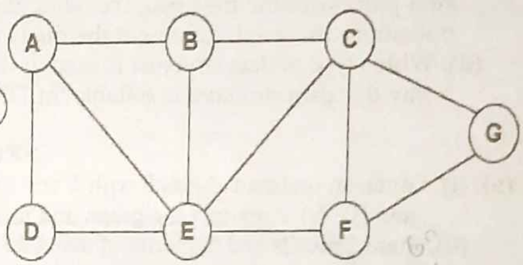
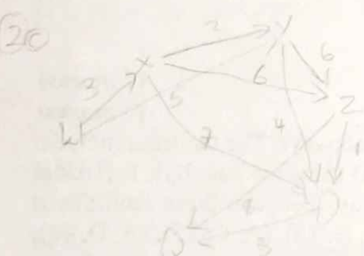
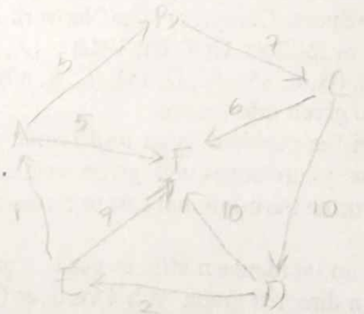


Figure 2





W → X → Y → Z → D → Q = 15

W 3→ X 3→ D → O = 13

W 2→ Y 4→ D 3→ O = 15

~~3 X 2 D 3 13~~

W 3→ X 6→ Z 8→ Q = 17

W 3→ X 3→ Y 5→ Z 3→ Q = 19

W 5→ Y 6→ Z 8→ Q = 17

BFS → FIFO V and Q

DFS → LIFO Stack

Dijkstra → all in one table

Bellman-ford – Iterations

Prim – tree